Nathaniel Powell

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Available for an 8-month co-op beginning January 2027

Education

University of Waterloo — Waterloo, ON

Bachelor of Science, Honours Mathematical Physics, Co-op — *Sep. 2025 – Present*

Coursework and experience spanning applied mathematics, physics, programming, and computational problem solving.

Technical Skills

Languages: Python, Java, JavaScript, SQL
Web & Application Development: React, Node.js, Supabase/PostgreSQL, REST APIs, OpenCV, Tkinter
Software Engineering: System Design, Data Modeling, Requirements Gathering, Performance Optimization, Testing, Debugging, Object-Oriented Programming, Multithreading
Tools: Git, GitHub, Figma, Microsoft Excel, PowerPoint

Experience

Quadmates.ca — Waterloo, ON

Founder & Developer — *2026 – Present*

- Designed, developed, and deployed a **full-stack web application** using React, Node.js, JavaScript, and Supabase/PostgreSQL, supporting approximately **50 early users**.
- Designed application architecture, database schemas, APIs, and **end-to-end data flows** across frontend, backend, authentication, and persistent storage layers.
- Performed **performance optimization** across direct and group messaging workflows, reducing latency by **70%** and improving responsiveness of real-time application features.
- Built and tested **3 end-to-end workflows** spanning authentication, student profiles, and study-partner matching, troubleshooting issues across frontend, backend, API, and database components.
- Integrated **AI-assisted development workflows** into debugging, code generation, refactoring, and feature implementation to accelerate development and resolve technical issues.
- Collected user feedback and translated **user and business requirements into technical requirements**, prioritizing features and refining system functionality throughout development and launch.

UW Orbital — Waterloo, ON

Business & Finance Team Member — *Jan. 2026 – Present*

- Developing a **Python automation workflow** to streamline expense-data processing and reduce manual updates to financial records.
- Built and maintain an **Excel-based expense dataset** to centralize spending information, organize financial records, and support budget tracking.
- Collaborate within a **406-member multidisciplinary engineering organization** supporting satellite design and development.
- Conducted external stakeholder outreach to a company of approximately **50 employees** to pursue fabrication materials and engineering support.

Projects

Computer Vision Security Camera | *Python, OpenCV, Tkinter*

- Developed a real-time **Python computer-vision application** using OpenCV HOG/SVM detection to identify people from live camera input.
- Implemented a **multithreaded producer-consumer pipeline** using a 5-frame queue and every-third-frame processing to manage computational load.
- Integrated live video processing and audible alerts into a desktop application for real-time monitoring.